

Milestone 02

Answer Key

Environment

Update the YAML with your first and last name.

Load the packages.

```
library(tidyverse)
library(haven)
library(gssr)
library(gssrdoc)
library(summarytools)
```

Load the 2024 GSS data and the GSS panel data

```
# ADD YOUR CODE HERE TO LOAD THE GSS 2024 DATA
gss24 <- gss_get_yr(2024)
```

You'll be working with the following GSS variables:

- **hrs1**: Hours worked per week
- **tvhours**: Hours spent watching TV per day
- **sex**: Respondent's gender
- **polviews**: Political views (from extremely liberal to extremely conservative)

Code, Output, Meaning

Use R to complete the checkpoints below. Show your work (e.g., R code chunks) where appropriate. Add narrative (text outside code chunks) or comments (text inside the code chunks) throughout.

Reference specific statistics (where appropriate) from your output to justify your answers. Explain what the values tell you about the data; interpret their meaning in relation to the question.

Checkpoint 01: Clean variables

Use `zap_missing()`, `as_factor()`, and `droplevels()` to make `sex` into a clean factor variable.

```
gss24$sex <- zap_missing(gss24$sex)
gss24$sex <- as_factor(gss24$sex)
gss24$sex <- droplevels(gss24$sex)
```

Checkpoint 02: Summarize weekly work hours (`hrs1`)

Use `drop_na` and `summarize()` to create a table showing the frequency, median, mean, and sd of `hrs1`

BONUS: use `round()` to indicate keeping 2 decimal places.

```
gss24 |>
  drop_na(hrs1) |>
  summarize(
    freq = n(),
    median_hrs1 = round(median(hrs1), digits = 2),
    mean_hrs1 = round(mean(hrs1), digits = 2),
    sd_hrs1 = round(sd(hrs1), digits = 2)
  )
```

```
# A tibble: 1 x 4
  freq median_hrs1 mean_hrs1 sd_hrs1
<int>      <dbl>     <dbl>   <dbl>
1  1768         40      39.4    13.9
```

Is the median value for `hrs1` consistent with the mean? Speculate what might explain any differences.

Interpret the mean and standard deviation for `hrs1`.

Checkpoint 03: Summarize daily tv hours (tvhours)

Use `drop_na()` and `summarize()` to create a table showing the frequency, median, mean, and sd of `tvhours`.

BONUS: use `round()` to indicate keeping 2 decimal places.

```
gss24 |>
  drop_na(tvhours) |>
  summarize(
    freq = n(),
    median_tv = round(median(tvhours), digits = 2),
    mean_tv = round(mean(tvhours), digits = 2),
    sd_tv = round(sd(tvhours), digits = 2)
  )
```

```
# A tibble: 1 x 4
  freq median_tv mean_tv sd_tv
<int>      <dbl>   <dbl> <dbl>
1  2152          2     3.3   3.3
```

Is the median value for tvhours consistent with the mean? Speculate what might explain any differences.

Interpret the mean and standard deviation for `tvhours`.

Checkpoint 04: Summarize daily tv hours (tvhours) by sex

Use `drop_na`, `group_by()` and `sumamrize()` to create a table showing the mean and sd of `tvhours` by sex.

```
gss24 |>
  drop_na(tvhours, sex) |>
  group_by(sex) |>
  summarize(
    mean = round(mean(tvhours), digits = 2),
    sd = round(sd(tvhours), digits = 2)
  )
```

```
# A tibble: 2 x 3
  sex      mean    sd
  <fct>   <dbl> <dbl>
1 male    3.21  3.24
2 female  3.38  3.35
```

How did the mean and standard deviation of daily tv hours differ between men and women and what do these differences suggest about gender-based variability in tv viewing patterns?

Checkpoint 05: Create a new data frame

Create a new data frame called `my_data` that contains only the four variables listed above. (Hint: use `select()`).

Then, use `head()` to preview `my_data`.

```
my_data <- gss24 |>
  select(hrs1, tvhours, sex, polviews)

head(my_data)
```

```
# A tibble: 6 x 4
  hrs1      tvhours      sex polviews
  <dbl+lbl> <dbl+lbl>   <fct> <dbl+lbl>
1    43         15   male      4 [moderate, middle of the road]
2 NA(i) [iap] NA(n) [no answer] male      3 [slightly liberal]
3 NA(i) [iap] NA(i) [iap]   female     1 [extremely liberal]
4    20          2   male NA(d) [don't know]
5 NA(i) [iap] NA(i) [iap]   female     4 [moderate, middle of the road]
6 NA(i) [iap]     2   male NA(s) [skipped on web]
```

Checkpoint 06: Remove rows with missing data

Use `drop_na()` to keep only respondents who have non-missing values for all variables in `my_data`.

Then, use `head()` to preview `my_data`.

```
my_data <- my_data |>
  drop_na()

head(my_data)
```

```
# A tibble: 6 x 4
  hrs1    tvhours    sex  polviews
<dbl> <dbl> <fct> <dbl>
1 43      15    male    4 [moderate, middle of the road]
2 80       1   female    2 [liberal]
3 35       4   female    2 [liberal]
4 40       2   female    1 [extremely liberal]
5 60       0 [0 hours] female    4 [moderate, middle of the road]
6 37       2   female    3 [slightly liberal]
```

Checkpoint 07: Create variables

A. Use `mutate()` and `case_when()` to add a new variable to `my_data` called `work_category` that groups respondents as follows:

- “Part-time” if `hrs1` is equal to or more than 0 and equal to or less than 34
- “Full-time” if `hrs1` is 35 or more

BONUS: use `table()` to show the counts of your new variable.

```
my_data <- my_data |>
  mutate(
    work_category = case_when(
      hrs1 >= 0 & hrs1 <= 34 ~ "Part-time",
      hrs1 >= 35 ~ "Full-time"
    )
  )

# table(my_data$hrs1, my_data$work_category)
table(my_data$work_category)
```

```
Full-time Part-time
      872      260
```

B. Use `mutate()` and `case_when()` to add a new variable to `my_data` called `pol3cat` that groups respondents as follows:

- “Liberal” if `polviews` equals “extremely liberal”, “liberal”, or “slightly liberal”
- “Moderate” if `polviews` equals “moderate, middle of the road”
- “Conservative” if `polviews` equals “extremely conservative”, “conservative”, “slightly conservative”

BONUS: use `table()` to show the cross-tab of `pol3cat` and `polviews`

```
my_data <- my_data |>
  mutate(
    pol3cat = case_when(
      polviews >= 1 & polviews <= 3 ~ "Liberal",
      polviews == 4 ~ "Moderate",
      polviews >= 5 & polviews <= 7 ~ "Conservative"
    )
  )

table(my_data$pol3cat, my_data$polviews)
```

	1	2	3	4	5	6	7
Conservative	0	0	0	0	139	179	50
Liberal	61	158	131	0	0	0	0
Moderate	0	0	0	414	0	0	0

Checkpoint 08: Summarize tv hours by work category

Use `group_by()` and `summarize()` to create a table showing the frequency, mean, median, and sd of `tvhours` for each of the two work-hours groups.

BONUS: use `round()` to indicate keeping 2 decimal places.

```
my_data |>
  group_by(work_category) |>
  summarize(
    count = n(),
    mean_tvhours = round(mean(tvhours), digits = 2),
    median_tvhours = round(median(tvhours)),
    sd_tvhours = round(sd(tvhours), digits = 2)
  )
```

A tibble: 2 x 5

work_category	count	mean_tvhours	median_tvhours	sd_tvhours
---------------	-------	--------------	----------------	------------

	<chr>	<int>	<dbl>	<dbl>	<dbl>
1	Full-time	872	2.55	2	2.45
2	Part-time	260	2.77	2	2.69

Which work-hours group reported the most daily tv hours in 2024? Was there a lot or a little variability in tv hours?

Is the median value for tvhours consistent with the mean? Speculate what might explain any differences.

Checkpoint 09: Summarize weekly work and daily tv hours by political identity

- A. Use `group_by()` and `summarize()` to create a table showing the frequency, mean, median, and sd of `hrs1` for each of the three political identity groups.

BONUS: use `round()` to indicate keeping 2 decimal places.

```
my_data |>
  group_by(pol3cat) |>
  summarize(
    count = n(),
    mean_hrs1 = round(mean(hrs1), digits = 2),
    median_hrs1 = round(median(hrs1)),
    sd_hrs1 = round(sd(hrs1), digits = 2)
  )
```

```
# A tibble: 3 x 5
  pol3cat      count mean_hrs1 median_hrs1 sd_hrs1
  <chr>      <int>     <dbl>     <dbl>    <dbl>
1 Conservative   368      39.9        40     14.7
2 Liberal        350      38.4        40     13.4
3 Moderate       414      39.4        40     13.8
```

Which political group reported the most work hours in 2024? Was there a lot or a little variability in work hours?

Is the median value for hrs1 consistent with the mean? Speculate what might explain any differences.

- B. Use `group_by()` and `summarize()` to create a table showing the frequency, mean, median, and sd of `tvhours` for each of the three political identity groups. **BONUS:** use `round()` to indicate keeping 2 decimal places.

```
my_data |>
  group_by(pol3cat) |>
  summarize(
    count = n(),
    mean_tv = round(mean(tvhours), digits = 2),
    median_tv = round(median(tvhours)),
    sd_tv = round(sd(tvhours), digits = 2)
  )
```

```
# A tibble: 3 x 5
  pol3cat      count mean_tv median_tv sd_tv
  <chr>      <int>   <dbl>    <dbl> <dbl>
1 Conservative   368     2.56         2  2.71
2 Liberal        350     2.47         2  2.3
3 Moderate       414     2.75         2  2.49
```

Which political group watched the most television in 2024? Was there a lot or a little variability in television hours?

Checkpoint 10: Create a summary dataframe

Use `group_by()` and `summarize()` to create and save a dataframe showing the average work (name it `avg_work_24` and tv hours (name it `avg_tv_24`) for men and women in 2024.

```
gss24_summary <- my_data |>
  group_by(sex) |>
  summarize(
    avg_work_24 = round(mean(hrs1), digits = 2),
    avg_tv_24 = round(mean(tvhours), digits = 2)
  )

head(gss24_summary)
```

```
# A tibble: 2 x 3
  sex      avg_work_24 avg_tv_24
  <fct>      <dbl>    <dbl>
1 male         41.7        2.53
2 female       37.0        2.67
```


Did men or women report more average work hours in 2024? What about TV hours?

What percentage of men worked 40 hours or less than per week in 2024? (You can either add R code chunk to this document or calculate it with the formula. Show your work either way.)

IPUMS Data

To keep your Research Brief progress on track, you'll complete short exercises that correspond with the new course material using your own dataset as part of your milestones.

- Present a (pretty) relative frequency table of a nominal or ordinal variable in your dataset.
- Create a summary table for a key interval-ratio (e.g., continuous) variable you are interested in exploring more in your dataset. Include the frequency, mean, median, and sd.

Some potentially helpful resources:

- [package `ipumsr\(\)`](#)
- [Webinar: Using IPUMS data in R with `ipumsr`](#)
- [Just the slides: Using IPUMS data in R with `ipumsr`](#)